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METHOD OF SECURE DATA STORAGE AND TRANSFER

Abstract

A method of secure data transfer and storage using a storage device storing encrypted information. The method uses a host that stores and transfers encrypted sensitive information and a customer that desires the information to be securely stored. The customer chooses a unique encryption code to encrypt sensitive information and uploads the information to a host. Then the host transfers the information to the storage device. Connections between the host and customer are intermittent and done so the storage device remains isolated from networks outside of the host.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation-in-Part and claims the benefit of priority from U.S. patent application Ser. No. 17/457,532, filed Dec. 3, 2021, which claims the benefit of U.S. Utility patent application Ser. No. 16/711,650, filed on Dec. 12, 2019, and the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] This present disclosure relates to information stored on computer disks, USB devices, and removable storage. There are multiple ways data can be destroyed, compromised, or damaged. Data can be destroyed through hardware failure, such as a hard drive crash or power surge, or through fire, flood, or mechanical force. Data stored on networked computers can also be compromised through hacking over the internet. It can also be compromised by careless behavior such as leaving a computer logged in and unattended, leaving an unsecured laptop out where it can get stolen, or losing a USB flash drive with sensitive information stored on it. Some systems and methods attempt to solve these problems, but often fall short. There are encryption algorithms to prevent a loose flash drive from being read, antivirus software to reduce the chances of a cyberattack, and enclosures to prevent destruction from fire, flood, or mechanical damage. Each system has individual vulnerabilities. An improved method is needed.

SUMMARY OF THE INVENTION

[0003] The present disclosure describes a method of using an air-gapped computer system in concert with ruggedized, encrypted on-site storage and portable encrypted storage. The method involves transfer of information from a customer's location to a secure location. The files are encrypted at the customer's location and sent over a secure connection. An intermittently gapped host computer is selectively isolated from any network outside of the host and can receive information from the customer over the secure connection. The intermittently gapped host computer may be connected to a storage device only when the intermittently gapped host computer is isolated from any network outside of the host. File transfer occurs between the on-site storage and removable storage only on the air-gapped computer system. The on-site storage retains previous versions of information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] A preferred embodiment of this invention has been chosen wherein:

[0005] FIG. 1 is a diagram of the system at the host;

[0006] FIG. 2 is an overall diagram of the systems at the host, customer, and the flow of information between them;

[0007] FIG. 3 is a flow diagram for when the customer sends sensitive information to the host;

[0008] FIG. 4 is a block diagram of how data is transferred when an internet connection is used with an intermittently gapped host computer; and

[0009] FIG. 5 is a block diagram of how data is transferred when two factor authentication is used.

DESCRIPTION OF THE PREFERRED EMBODIMENT

[0010] An apparatus and method of securely storing and updating information is described herein, involving a host **20**, a customer **22**, and the physical transportation of a removable device **24**. The removable device **24** is made to withstand rough handling and may incorporate ruggedizing materials and manufacturing. For example, the removable device **24** may be integrated into a durable and/or resilient enclosure. The enclosure may include padding, other mechanical shock protection, and/or protection from moisture. The host **20** is the secured location where sensitive

information is stored and archived. The customer **22** is located separately from the host **20** and has data that the customer desires to be securely backed up. Sensitive information is transferred between the host **20** and customer **22** completely offline, meaning the mail, hand-delivery, courier, or other method that is completely separate from any computer network. All file transfer and storage outside of the customer is encrypted and inaccessible without a customer-chosen unique encryption key **72**. Any intercept of the removable device **24** by an unauthorized user will result in unreadable and inaccessible information without the encryption key **72**.

[0011] The host **20** is a secure location and has a connected area **26** and an air-gapped area **28**. While they are shown adjacent, the connected and air-gapped areas **26**, **28** may be separate buildings, locations, or rooms. The connected area **26** is connected to the internet **38** but may have restricted access. The restricted access may include physically restricting access and/or restricting computer and network access. The air-gapped area **28** is completely separate from any outside communications and is surrounded by an “air-gap” **30**, meaning there is no internet, intranet, or other network connectivity to any devices contained inside the air-gapped area **28**. The air-gapped area **28** is protected from outside unauthorized access at the host **20**. The protection may be in the form of lock and key, key cards, physical keys, access codes, RFID tags, or other systems to restrict physical access to the air-gapped area **28**. Located inside the air-gapped area **28** is an array **33** of armored storage units **34**, each having their own barcoded serial number **36** or other unique identification. The armored storage units **34** described herein are metal enclosures with extreme heat-resistant insulation and components, but it is contemplated that other types of storage devices are used, such as removable hard drives or other storage media. It is contemplated that the armored storage units **34** are stored in fire resistant enclosures. The air-gapped area **28** may be supplied by an independent power source, backup generator, or otherwise be connected to an off-grid supply of power. The air-gapped area **28** further contains an air-gapped host computer **32** that may be located adjacent the array **33** or in a separate location. The air-gapped host computer **32** is completely standalone from any computer network, including any computer network at the host. Further, the air-gapped host computer **32** may contain a custom operating system, software, or firmware that is used for copying and updating sensitive information between the storage unit **34** and removable storage device **24**. The storage units **34** are normally in a disconnected state and reside in the storage array **33**, where no electricity is present and no communication to the storage unit **34** can occur. The storage units **34** are rugged and are protected from high heat, mechanical damage, liquid intrusion, and electrical events. Each storage unit **34** has a rigid metal shell with a tight-fitting lid. Inside the metal shell is high-temperature insulation that surrounds a storage unit. The storage unit may include phase-change heat absorbers that surround a solid-state storage device. Sacrificial wiring connects the solid-state storage device through a small gap in the rigid metal shell. In the event that there is a fire, flood, earthquake, collapse, or other damaging event inside the air-gapped area **28**, each storage unit **34** is individually protected. This is in contrast to a server or drive array that relies on protection supplied by the room or area where it is installed. In a server or drive array, a flood or fire could easily cause destruction of all data and hardware.

[0012] The connected area **26** has an internet-connected host computer **40**. For simplicity, the internet-connected host computer **40** is shown and described as a single computer, but the internet-connected host computer **40** can be a network or series of computers that may be completely located at the host or hosted off-site. There is no communication between the internet-connected host computer **40** and the air-gapped host computer **32**. The internet-connected host computer **40** has a barcode printer **42**, which is located at the host **20**. The internet-connected host computer **40** may have a database that stores customer information, such as the mailing address, billing information, file transfer log, serial number/ID of their storage unit **34**.

[0013] The customer has its own computer **50** or computer network that uses sensitive information. The sensitive information may be stored on a network drive, attached storage, or on the computer(s) directly. The customer's computer **50** may be connected to the internet **38** or be on an

air-gapped network. Sensitive information can include trade secrets, banking information, programs, personnel records, accounts and passwords, databases, or any type of electronic files or information that may be important to the customer. The term “sensitive information” is used as a definition for the files that are stored and transferred between the host and customer and is not limiting to the types of files that are stored, encrypted, and transferred. While the files are commonly sensitive or contain sensitive information, the files can also be ordinary and/or publicly-available.

[0014] An app **60** may be provided by the host that the customer installs on a device **62** that is connected to the internet **38**. This is commonly a smartphone, tablet, or other mobile device. The app **60** communicates with the connected host computer **40** through the internet **38** and is used to initiate the offline transfer of sensitive information. The app **60** also is used to track any shipment/delivery of the removable storage device **24**. Alternatively, the customer **22** may contact the host computer **40** through the internet via web portal, phone call, or email. The customer installs encryption software **70** and chooses a unique encryption key **72** that is not shared with anyone at the host **20**. Without the encryption software **70** and key **72**, the sensitive information is completely inaccessible, even by the host **20**. If the customer **22** loses the encryption key **72**, the data is inaccessible.

[0015] The air-gapped host computer **32** may include a barcode reader **74** to document and ensure that the barcodes **36**, **44** match before any file transfer. Because the files are not readable by the host, ensuring matching serial numbers prevents errors and loss of information.

[0016] There are different scenarios when transferring sensitive information. These include, but are not limited to, the following: initial load, updating, and restoration. In the initial load, the sensitive information is only at the customer's computer **50** and no sensitive information is present at the host **20**. When updating, the sensitive information at the customer's computer **50** has been updated or changed, and the customer desires the updated sensitive information to be stored at the host. During restoration, the sensitive information has been compromised or destroyed and the sensitive information stored at the host **20** is the only trusted copy.

[0017] Alternatively, the customer **22** desires to move it to a new computer or network. In all scenarios, the sensitive information never travels between the host **20** and customer **22** via a computer network or the internet **38**. The files are only transferred by physical delivery.

[0018] For the initial load, the customer **22** contacts the host for a blank removable device **24**. The connected host computer **40** prints out a unique barcode **44** that is tied to that customer and a designated storage unit **34**. The unique barcode **44** matches on the removable device **24** and storage unit **34**. The barcode **44** is applied to the blank removable device **24** and then the host **20** places the blank device **24** in the mail **80** to the customer. The mail **80** is tracked by the host **20** to ensure safe delivery to the customer **22**. The customer **22** installs encryption software **70** and chooses an encryption key **72**. If the encryption key **72** is lost, any encrypted information is inaccessible. Upon receipt of the removable device **24**, the customer **22** connects the blank removable device **24** to the customer computer **50** and enters the encryption key **72**. The customer **22** encrypts and loads all sensitive information onto the removable device **24**. This takes the sensitive information and transforms it to encrypted sensitive information as it is being loaded onto the removable device **24**. Upon completion, the customer **22** then mails **82** the removable device **24** to the host **20**. The host **20** tracks the package, either through the app **60** or being otherwise notified by the customer **22**. The sensitive information on the removable device **24** is encrypted. When the removable device **24** arrives at the host **20**, the barcode **44** is scanned at the internet-connected host computer **40** to tie it to the customer's account. The removable device **24** is never connected to the internet-connected host computer **40**, only the air-gapped host computer **32**. The host **20** identifies the correct armored storage unit **34** in the air-gapped area **28**. The armored storage unit **34** and removable device **24** are connected to the air-gapped computer **32** and the encrypted files are copied from the removable device **24** to the armored storage unit **34**. After file transfer is complete, the armored storage unit **34**

is disconnected and placed back in the array 33. The host 20 cannot access files on either device 24, 34 due to the encryption. The removable device 24 is then wiped clean of information and mailed 80 back to the customer 22. The customer 22 may be notified through the app 60 that the transfer is complete and the blank removable device 24 is on its way back to the customer 22.

[0019] To update the sensitive information, the customer 22 takes the removable device 24 and loads encrypted sensitive information onto the removable device 24. The customer 22 then notifies the host 20 through the app 60 and mails 82 the removable device 24 to the host 20. When the host 20 receives the removable device 24, the encrypted files are copied from the removable device 24 to the armored storage unit 34. A new folder is added to the armored storage unit 34 with the updated information. Each time the sensitive information is updated, a new folder is created to allow previous versions of sensitive information be retrievable by the customer in the event one version of the sensitive information is corrupted or otherwise compromised. It is contemplated that alternate methods of saving and retrieving versions of sensitive information are used.

[0020] To restore sensitive information, the customer 22 contacts the host 20 through the app 60. The host 20 may either load a new removable device 24 with sensitive information or mail 80 the armored storage unit 34 back to the customer 22. As with other mailing or shipping, the progress is tracked through the app 60.

[0021] In some instances, sensitive information may be transferred between the host 20 and the customer 22 through a secure online connection. This is shown in FIG. 4. In this case, the host 20 is located in a secure location and has an intermittently connected host computer 90 located with the host 20. The intermittently connected host computer 90 is connected to the internet 38 via a secure connection 39. The secure connection 39 may include SSL. The secure connection 39 to the internet 38 is between a device 98 located with the customer 22 and the intermittently connected host computer 90. The information leaving the device 98 is encrypted 100 before it leaves the customer 22. The device 98 at the customer 22 may be a computer or local cloud and may even be a network of computers. Once the information is encrypted 100, that information may only be viewed by using an encryption key 72 that the customer 22 retains. In addition to the information being encrypted before it leaves the customer 22, the information may be encrypted at a server on the internet 38 between the device 98 located with the customer 22 and the intermittently connected host computer 90. Thus, any information that may be potentially intercepted on the internet 38 would be unreadable without the key 72 and that is still true if the information were decrypted during its time on the internet 38 because such decryption would only be for the second layer of encryption. The device 98 may include special software that strictly limits the amount of time information is sent via the internet 38. For instance, the software on the device 98 may intermittently transmit information through the internet 38 by only activating a connection to the internet 38 on a predetermined time interval. The software may be set up to send information on the internet 38 once every five minutes in a burst of packets. Such transmissions periodically limit the amount of time the information spends in transit on the internet 38 and minimizes the time it may be intercepted. The user of the device 98 may select the interval of each transmission of information to a longer or shorter time. Such a choice for the predetermined time interval before sending information via the internet 38 would depend on the user and a shorter time interval before sending information would limit the amount of time information is not synchronized with the intermittently host computer 90. In addition to the user limiting the amount of time information is transmitted over the internet 38 at the device 98, the intermittently connected host computer 90 may do so as well. Before connecting to the intermittently connected host computer 90, a login is required to make that connection.

[0022] The login may be monitored through a firewall 51 between the device 98 and the intermittently connected host computer 90. To add a further layer of security, the intermittently connected host computer 90 may have a switch 99 to affirmatively break its connection to the internet 38 and provide the intermittently connected host computer 90 an air-gapped connection.

The firewall **51** is connected to the switch **99** such that the firewall **51** can prevent the switch **99** from closing. When the device **98** attempts to login to the intermittently connected host computer **90** to make a secure connection, the host **20** sends a unique identifier code **55** to the customer **22**. The host **20** can send this unique identifier code **55** to the customer **22** in number of different ways, including through the internet **38**. The host **20** can send the unique identifier code **55** to the same device which the customer **22** is attempting to securely connect to with the intermittently connected host computer **90**. This unique identifier code **55** can be sent to the customer **22** through a text or email to the same device **98** or a different device. The host **20** also ensures that the firewall **51** has a copy of the unique identifier code **55** sent to the customer **22**. The login directs the customer **22** to enter a code. The firewall **51** compares the code received from the customer **22** to the code received from the host **20**. If the code entered by the customer **22** into the input box does not match the unique identifier code **55**, the firewall **51** does not allow the switch **99** to close and the customer **22** cannot connect to the intermittently connected host computer **90**. When the customer **22** enters the unique identifier code **55** into the input box, the firewall **51** does not prevent the switch **99** from closing. Thus, the firewall **51**, along with the login, ensures that the connection between the host **20** and the customer **22** is a secure connection confirmed by multiple forms of identification. The customer **22** must use the correct login and must match the unique identifier code **55** in the input box before the intermittently connected host computer **90** can connect to the device **98** through the internet **38**. The firewall-protected switch **99** is never closed at the same time as the switch **122** between the intermittently connected host computer **90** and the armored storage unit **34**. This ensures that the air-gapped area **28** remains truly air gapped and secure. The firewall **51** can never be used to over-ride the other safeguards so that the switch **99** and switch **122** are closed at the same time. The firewall **51** ensures that the switch **99** remains open and the intermittently connected host computer **90** remains disconnected from the device **98** through the internet **38**. In this way, the firewall **51** ensures that the intermittently connected host computer **90** only connects to the customer's device **98** and not some other device. Because the firewall **51** and switch **99** prevent the intermittently connected host computer **90** from sending or receiving packets of information through the internet **38** to any device other than the customer's device **98**, the connection is secure.

[0023] The intermittently connected host computer **90** also has the capacity to monitor activity from a customer **22** connecting to it or attempting to connect to it. In the absence of activity, the intermittently connected host computer **90** could time the customer **22** out after a predetermined time thereby breaking the connection to the internet **38** to minimize the time the intermittently connected host computer **90** and the device **98** are connected.

[0024] The intermittently connected host computer **90** retains a serial number **110**, or other unique identification, that is provided to the customer **22** and associated with an armored storage unit **34**. The associated serial number **110** of the armored storage unit **34** is also associated with a particular customer **22** and is stored in an array **33** of armored storage units **34**. As described, the array **33** of armored storage units **34** are stored in an air-gapped area **28**. Each armored storage unit **34** is an extremely durable unit that has the capacity to survive many types of disasters including fire, flood, or most types of damaging event that would destroy a less durable storage device.

[0025] The intermittently connected host computer **90** has an intermittent connection **118** to the customer **22** and also has an intermittent connection **118** to storage units **34** that correspond to particular customers **22**. An intermittent connection **118** is made to each storage unit **34** within the array **33**. The intermittent connection **118** is gapped via the use of a switch **122** or other gapping device that does not leave the storage unit **34** connected while the intermittently connected host computer **90** is connected to the customer **22** via the internet **38**. The switch **122** may be a relay or other type of switch that is electrically controlled to physically break the connection **118** between the intermittently connected host computer **90** and the storage units **34** within array **33**. Each armored storage unit **34** must be connected to the intermittently connected host computer **90**

through a switch **118** and although only one switch **118** is shown, there may be many switches connecting the intermittently connected host computer **90** to each storage unit **34** within the array **33**. In this manner, there is a physical break between each storage unit **34** and the world outside of the air-gapped area **28**. Generally, for a fully gapped air-gapped area **28**, the switches **122** to any one of the storage units **34** in the air-gapped area will not be connected while the intermittently connected host computer **90** is connected to the internet **38**. In other words, if the connection to the internet **38** is viewed as a switch **122** that is closed when the intermittently connected host computer **90** is connected to the internet **38**, the switch **122** is open so that the air-gapped area **28** remains truly air gapped. More simply stated, the switch **122** will never be closed when the intermittently connected host computer **90** is connected to the internet **38**. The intermittently connected host computer **90** may be configured to store information from particular customers **22** it receives throughout a time period, then, for a select timeframe, deny all connections to the internet **38** so that switches **122** to particular storage units **34** corresponding to customers **22** who sent information may be transferred through the connection **118** into the air-gapped area **28**. In this manner, the air-gapped area **28** may be truly air-gapped because no computer or device connected to a storage unit **34** is connected to the internet **38** or any other network when information is being transferred to the storage unit **34**. As such, although the intermittently connected host computer **90** may at times be connected to the internet **38**, it is not connected to the internet **38** when it is connected to a storage unit **34**. Therefore, the intermittently gapped host computer is selectively isolated from any network to facilitate secure transfer of information to the storage units **34**.

[0026] To add a further layer of security, the intermittently connected host computer **90** may have a switch **99** to affirmatively break its connection to the internet **38** and provide the intermittently connected host computer **90** an air-gapped connection. The storage units **34** are always isolated from any network. The delay that the intermittently connected host computer **90** has while it holds information before being transferred to a corresponding storage unit **34** may be set to a predetermined time. If all information received via the internet **38** at the intermittently connected host computer **90** that is destined for a particular storage unit **34** is updated via connection **118** after the normal business hours of the host **20** during the night, a customer **22** seeking to retrieve his particular storage unit **34** the next morning can be assured that it will contain the information sent on the previous day. The general purpose of the network shown in FIG. **4** is to have a nearly continuously updated version of the device **98** located at the customer **22** to be replicated on the storage unit **34**. In addition to the storage unit **34** located at the host **20**, the customer **22** may also have a locally located storage unit **34'** that is the same as that in the array **33** at the host **20**. Such a locally located storage unit **34'** is continuously synchronized with the device **98**.

[0027] In addition to having a host **20**, there may be a second host **136** located remotely from the host **20**. The second host **136** may be a mirrored host having all of the same information stored within its array **33'** as in the array of the host **20**. The second host **136** may be connected through a direct connection **142** to the host **20**. This direct connection **142** may be separate from the internet **38** provided the second host **136** is in a location that provides the ability to have a direct connection **142** apart from the internet **38**. In other instances, the direct connection **142** to the second host **136** may be through the internet **38**. The second host **136** also has an intermittently connected computer **148** that connects to the array **33'** via a switch **150**. Switch **151** may be used to selectively disconnect the intermittently connected computer **148** from the internet **38**. In this configuration when switch **99** or **151** are closed, neither switch **122** nor **150** may be closed. This will maintain the air-gap and ensure storage units **34** are never connected to the internet **38**. Thus, the second host **136** may have an air gap **30'** surrounding air-gapped area **28'** that functions as the air-gapped area **28** at the host **20**.

[0028] It is understood that while certain aspects of the disclosed subject matter have been shown and described, the disclosed subject matter is not limited thereto and encompasses various other embodiments and aspects. No specific limitation with respect to the specific embodiments

disclosed herein is intended or should be inferred. Modifications may be made to the disclosed subject matter as set forth in the following claims.

Claims

1. A method of secure data transfer and storage between a host and a customer, said method comprising: said host providing an intermittently gapped host computer in a secure location, said intermittently gapped host computer selectively isolated from any network; said host providing a storage unit in said secure location, said storage unit isolated from any network outside of said host; providing a secure connection between said intermittently gapped host computer and a customer computer; serializing said storage unit; said customer attempting to make said secure connection to said intermittently gapped host computer and said host generating a code for said customer and sending said code to said customer; said customer communicating said code to said host and upon said customer matching said code from said host, said host allowing said secure connection; said customer encrypting and transferring sensitive information from said customer computer to said intermittently gapped host computer via said secure connection; disconnecting said intermittently gapped host computer from said secure connection; and said host transferring said encrypted sensitive information from said intermittently gapped host computer to said storage unit having a serial number corresponding to said customer while said intermittently gapped host computer is disconnected from said secure connection and any network outside of said host.
2. The method of secure data transfer in claim 1, wherein said secure connection is over an internet.
3. The method of secure data transfer in claim 1, wherein said storage unit is in an array including other storage units and said intermittently connected host computer is selectively connected to said storage unit.
4. The method of secure data transfer in claim 3, wherein said array is in an air-gapped area.
5. The method of claim 1, wherein a switch is located in a connection between said intermittently connected host computer and said storage unit, said switch open when said intermittently connected host computer is connected to said secure connection and said switch being closed only when said intermittently connected host computer is not connected to said secure connection.
6. The method of claim 5, wherein said secure connection to said intermittently connected host computer may be connected and disconnected with a switch.
7. A method of secure data transfer and storage for sensitive information comprising the steps of: providing an intermittently gapped host computer in a secure location, said intermittently gapped host computer selectively connectable to an internet connection; said host providing a storage unit in said secure location, said storage unit isolated from any network outside of said host; a customer computer being connectable to said intermittently gapped host computer via said internet connection; assigning a unique serial number to said storage unit; encrypting said sensitive information on said customer computer; said customer attempting to make said secure connection to said intermittently gapped host computer and said host generating a code for said customer and sending said code to said customer; said customer communicating said code to said host and upon said customer matching said code from said host, said host allowing said secure connection; connecting said customer computer to said intermittently gapped host computer via said internet connection and transferring said encrypted sensitive information to said intermittently gapped host computer; maintaining a database storing customer information and said unique serial number corresponding to said storage unit; disconnecting said intermittently gapped host computer from said internet connection and transferring said sensitive information from said intermittently gapped host computer to said storage unit only when said intermittently gapped host computer is disconnected from said internet connection.
8. The method of claim 7, wherein said intermittently connected host computer is selectively connected to said internet connection with a switch, disconnecting said switch providing an air-gap

when said switch is open.

9. The method of claim 7, wherein said storage unit is in an array including other storage units and said intermittently connected host computer is selectively connected to said storage unit.

10. The method of claim 9, wherein said array is in an air-gapped area.

11. The method of claim 7, wherein a switch is located in a connection between said intermittently connected host computer and said storage unit, said switch open when said intermittently connected host computer is connected to said internet connection and said switch being closed only when said intermittently connected host computer is not connected to said internet connection.
